Daily Geometry Drill #1

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Gradients & Perpendicularity · Distance & Midpoints · Perpendicular Distance · Shoelace Areas · Collinearity.

Section A Rapid Recognition

1. Find the gradient of the line $6x - 3y + 15 = 0$.

Ans: 2

2. Find the gradient of the line through $(1, 2)$ and $(5, 14)$.

Ans: 3

3. A line of gradient 4 is perpendicular to a line of gradient m. Write down m.

Ans: $-\frac{1}{4}$

4. Find the distance between $(2, 1)$ and $(5, 5)$.

Ans: 5

5. Find the midpoint of the segment joining $(4, 8)$ and $(10, 2)$.

Ans: "$(7, 5)$"

6. Find the perpendicular distance from the origin to the line $3x + 4y - 10 = 0$.

Ans: 2

7. A line has x-intercept 4 and y-intercept 6. Find its gradient.

Ans: $-\frac{3}{2}$

8. Find the area of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 3)$.

Ans: 6

9. The points $(1, 2)$, $(2, 4)$ and $(k, 10)$ are collinear. Find k.

Ans: 5

10. Using the shoelace formula, find the area of the triangle with vertices $(0, 0)$, $(3, 4)$ and $(6, 0)$.

Ans: 12

Section B Manipulation Drills

1. The line $2x - 3y + 6 = 0$ is perpendicular to which of the following?

A) $y = -\frac{3}{2}x + 1$

B) $y = \frac{3}{2}x - 2$

C) $y = \frac{2}{3}x + 5$

D) $y = -\frac{2}{3}x + 3$

Ans: A) $y = -\frac{3}{2}x + 1$

2. What is the perpendicular distance from $(2, 3)$ to the line $5x + 12y - 7 = 0$?

A) 1

B) 2

C) 3

D) 4

Ans: C) 3

3. Which point lies on the perpendicular bisector of the segment joining $(1, 2)$ and $(5, 6)$?

A) $(2, 5)$

B) $(3, 3)$

C) $(4, 4)$

D) $(5, 3)$

Ans: A) $(2, 5)$

4. Find the area of the quadrilateral with vertices $(0,0)$, $(4,0)$, $(5,3)$, $(1,3)$, in that order.

A) 10

B) 12

C) 14

D) 16

Ans: B) 12

5. The line through $(1,2)$ and $(4,5)$ crosses the y-axis at which value?

A) 0

B) 1

C) 2

D) 3

Ans: B) 1

6. Find the perpendicular distance from $(0, 1)$ to the line $6x - 8y + 2 = 0$.

Ans: $\frac{3}{5}$

7. Which of the following is NOT parallel to $y = 3x + 1$?

A) $y = 3x + 5$

B) $6x - 2y = 4$

C) $3x - y = 2$

D) $y = 2x + 1$

Ans: D) $y = 2x + 1$

8. The line $y = x + 1$ meets the line through $(0, 3)$ perpendicular to it at the point P. Find the coordinates of P.

Ans: "$(1, 2)$"

9. Which point is closest to the origin?

A) $(1, 3)$

B) $(2, 2)$

C) $(4, 0)$

D) $(0, 5)$

Ans: B) $(2, 2)$

10. Using the shoelace formula, find the area of the quadrilateral with vertices $(1, 0)$, $(5, 2)$, $(4, 6)$, $(0, 3)$, in that order.

Ans: $\frac{33}{2}$

Section C Substitution & Structure (≤90 s each)

Competition-hard: hidden structure, non-routine use of the same tools. No question is a direct formula plug.

1. The line $l : y = 6 - 2x$ meets the x-axis at A. A line m, perpendicular to l, meets the negative x-axis at $(k, 0)$. The lines l, m and the x-axis enclose a triangle of area 20. What is k?

A) -7

B) -10

C) $3 - 5\sqrt{2}$

D) $3 - 2\sqrt{5}$

Ans: A) -7

2. The perpendicular bisector of the segment joining $A(2, -6)$ and $B(b, 4)$, where $b > 0$, passes through the origin. What is b?

A) $2\sqrt{10}$

B) $2\sqrt{14}$

C) 12

D) $2\sqrt{6}$

Ans: D) $2\sqrt{6}$

3. The point $A(2, 9)$ is reflected in the line $x + 2y = 5$ to give the point A'. How far is A' from the origin?

A) $\sqrt{85}$

B) $\sqrt{10}$

C) 5

D) $\sqrt{185}$

Ans: C) 5

4. The lines $3x - 4y + 2 = 0$ and $4x - 3y - 5 = 0$ cross, dividing the plane into four angular regions. Which line bisects the angle whose region contains the point $(1, 1)$?

A) $x - y = 7$

B) $x + y = 7$

C) $7x + 7y = 3$

D) $7x - 7y = 3$

Ans: D) $7x - 7y = 3$

5. The points $A(1,4)$, $B(4,7)$ and $C(10,k)$ are collinear. The line through C perpendicular to AB meets the y-axis at D. What is the y-coordinate of D?

A) 23

B) 24

C) 25

D) 26

Ans: A) 23

6. Triangle ABC has vertices $A(0,0)$, $B(9,3)$ and $C(3,9)$. The point P lies on AB with $AP : PB = 1 : 2$, and Q lies on AC with $AQ : QC = 2 : 1$. What is the area of the quadrilateral $PBCQ$?

A) 24

B) 28

C) 27

D) 32

Ans: B) 28

7. The lines $y = 2x + 1$, $y = 7 - x$ and $y = \frac{1}{2}x - 2$ enclose a triangle. What is its area?

A) 12

B) 18

C) 24

D) 48

Ans: C) 24

8. The quadrilateral Q has vertices $(0, 0)$, $(4, 0)$, $(5, 3)$ and $(1, 3)$ in that order (a parallelogram). A second quadrilateral Q' is obtained by reflecting Q in the line $y = x$. What is the area of the overlap $Q \cap Q'$?

A) 6

B) 8

C) 9

D) 12

Ans: A) 6

Section D Challenge (TMUA / SMC / BMO1 difficulty)

Hardest 5: deep insight, hidden structure, or a construction you must discover. No diagram gives away the trick.

1. The points $A(1,3)$ and $B(7,5)$ are fixed, and P moves along the line $y = x$. What is the least possible value of $PA + PB$?

A) $4\sqrt{2}$

B) $2\sqrt{34}$

C) 10

D) $2\sqrt{10}$

Ans: D) $2\sqrt{10}$

2. The square $MNOP$ has side 10. Points R, S, T, U lie on MN, NO, OP, PM respectively, and $RSTU$ is a rectangle whose sides make $45°$ with the sides of the square. The area of $RSTU$ is 20. What is the length of a diagonal of $RSTU$?

A) $4\sqrt{15}$

B) $4\sqrt{10}$

C) 10

D) $4\sqrt{5}$

Ans: B) $4\sqrt{10}$

3. What is the area of the region of points (x, y) satisfying both $|x| + 2|y| \leq 6$ and $y \geq x - 2$?

A) 10

B) 18

C) 26

D) 36

Ans: C) 26

4. How many straight lines through the point $(2, 3)$ form, together with the two coordinate axes, a triangle of area 12?

A) 1

B) 2

C) 4

D) 3

Ans: D) 3

5. For each t with $0 < t < 4$, the line L_t passes through $(t, 0)$ and $(0, 4 - t)$. What is the greatest possible distance from the origin to L_t?

A) 1

B) 2

C) $\sqrt{2}$

D) $2\sqrt{2}$

Ans: C) $\sqrt{2}$